

TIC 72729777

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 28.675544 d (ALIASED, see below)
-> corrected period: 57.351087 d
Epoch (t0) 3938.8622 BTJD
Duration 0.1700 d
Depth 21.547 % (215,473 ppm)
SNR 1480.3

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

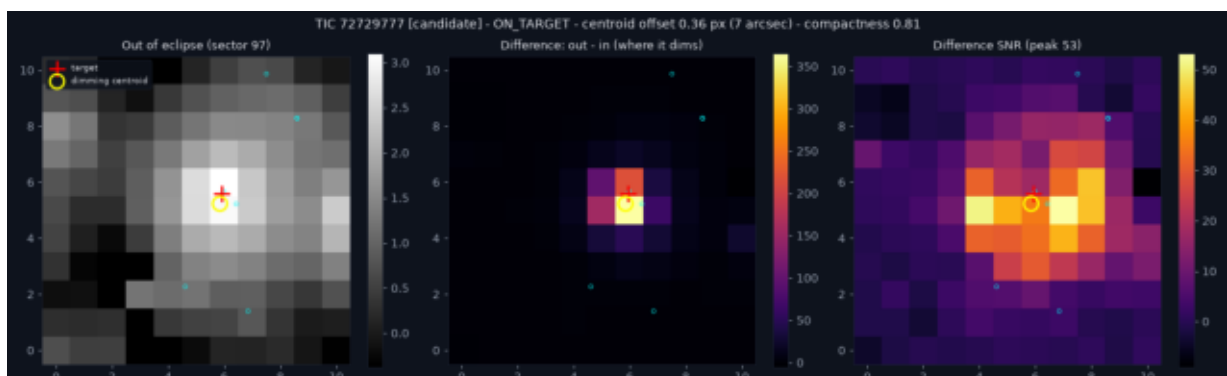
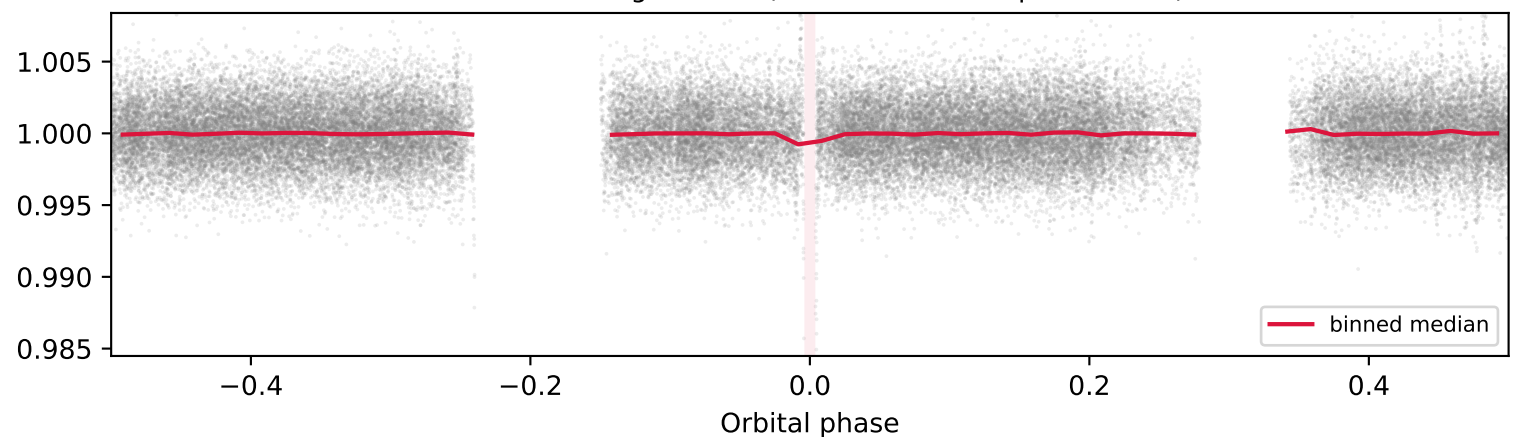
Eclipses observed 3 (gate ≥ 3)
Robust depth ratio 1.02 (gate ≥ 0.50)
Robust depth sigma 1103.1 (gate ≥ 7.0)
Odd/even z-score 111.74 (ALIAS FLAGGED)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 97 light-curve dip in this sector: 910.6 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 53.24 (gate ≥ 5.0) Compactness 0.81 (gate ≥ 0.50)
Dimming centroid offset 0.36 px (7 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: none within the cutout

Phase-folded light curve (shaded = BLS eclipse window)



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 111.74 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 57.351087 d, in any write-up.